CAD Ingestion Pipeline

Steps 1–5 · `src/pipeline/` · `python -m src.pipeline run` · 2026-10-27

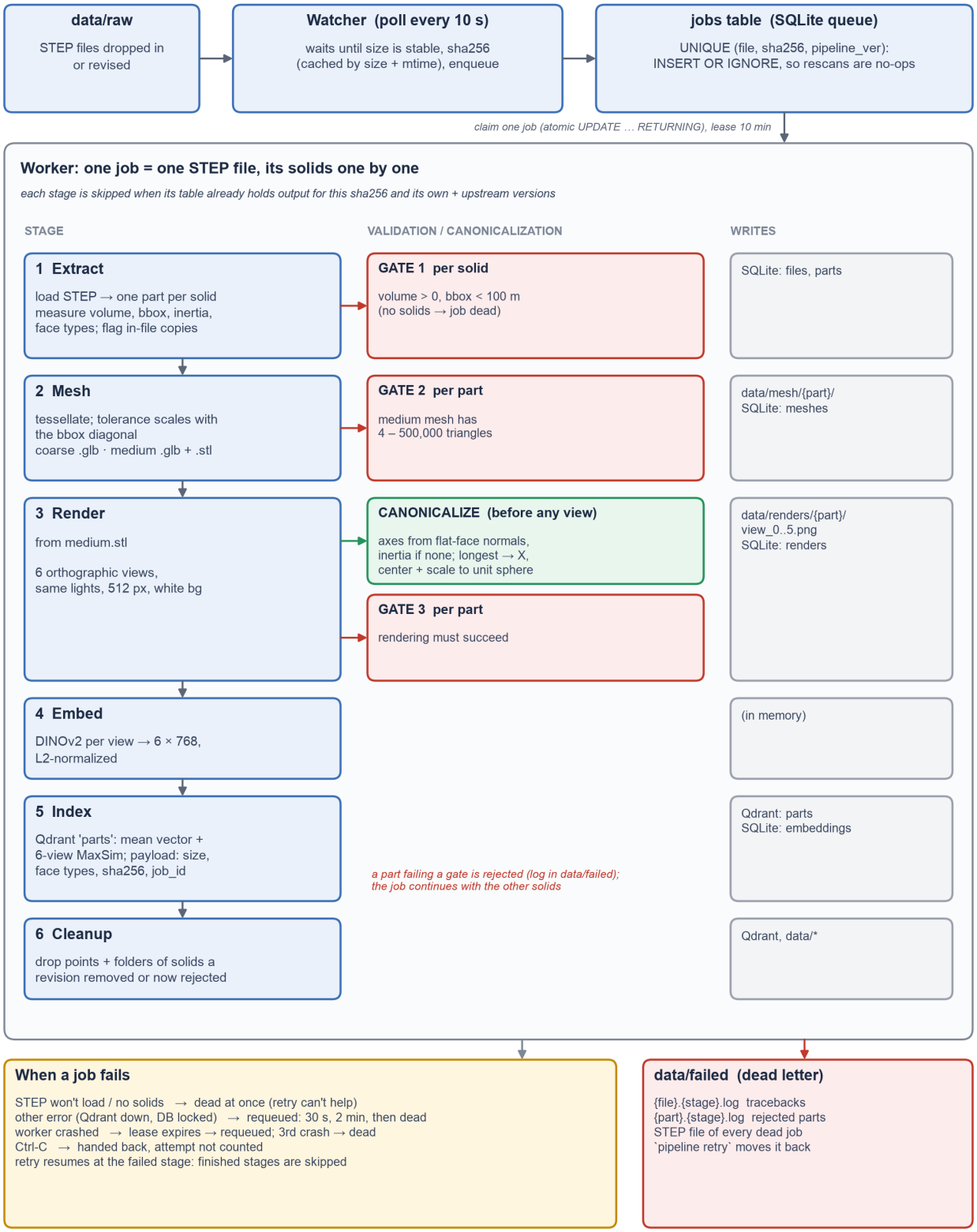


Figure 1. One STEP file through the pipeline. Red: validation gates (a failing part is rejected, the job continues). Green: canonicalization, inside render, before any view.

Stages

Stage	Output	Skipped when its table has
1 Extract	parts rows: volume, bbox, inertia, face types, duplicate flag	sha256 + EXTRACT_VER
2 Mesh	coarse.glb, medium.glb + .stl	sha256 + MESH_VER
3 Render	view_0..5.png, canonical pose	sha256 + MESH_VER + RENDER_VER
4–5 Embed, index	6 × 768 DINOv2 → Qdrant 'parts' (mean + MaxSim)	model + sha256 + MESH/RENDER_VER
6 Cleanup	drops what a revision removed	—

Skip keys include upstream versions, so a retry resumes at the failed stage and a version bump backfills only the changed stage and those after it.

Validation and canonicalization

- **Gate 1** (extract, per solid): volume > 0, largest bbox side < 100 m. No solids or unloadable STEP → job dead.
- **Gate 2** (mesh, per part): 4 – 500,000 triangles on the medium mesh. Caught a 2-triangle part.
- **Gate 3** (render, per part): all six views must render.
- **Canonicalize** (render, before views): axes from flat-face normals (inertia if none), longest → X, never mirrored; center + unit scale. Random-rotation search hits: 23–70% without, 100% with.

Jobs

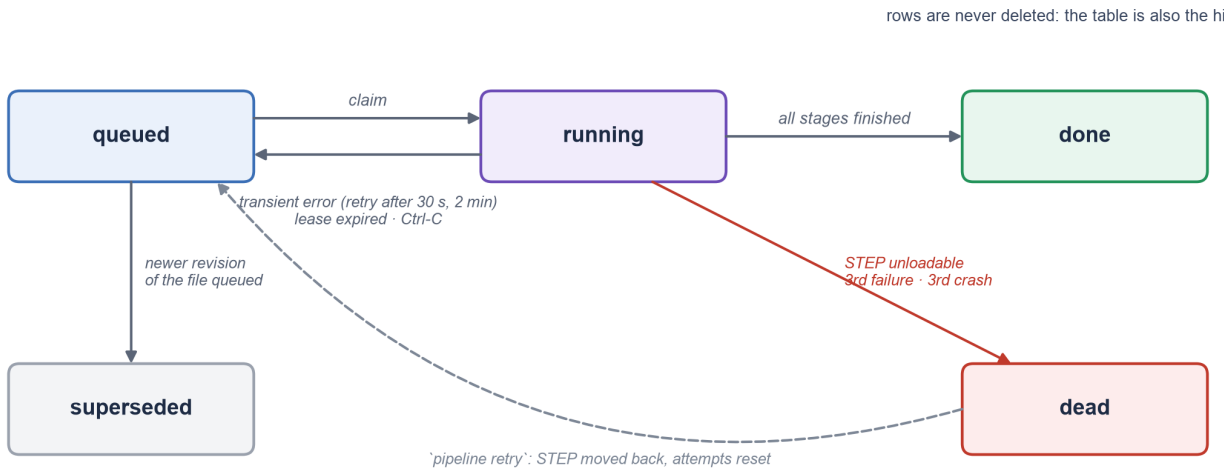


Figure 2. A job is one file at one sha256 and one pipeline version; UNIQUE (file, sha256, pipeline_ver) makes rescans no-ops.

Measured (500 files)

- 492 done, 10 dead; 1,625 searchable parts (67% of new solids were in-file duplicates).
- Drop → searchable ~16 s; 396 new files in 22.5 min; unchanged rerun 5 s.
- 1.6 GB total: mesh 699 MB, Qdrant 151 MB, renders 88 MB for 686 MB of STEP.

Known gaps

- Deleted files stay searchable; new parts get no PLM records until step 6 data is rebuilt.
- Batch `render` skips gate 2; one worker on SQLite.